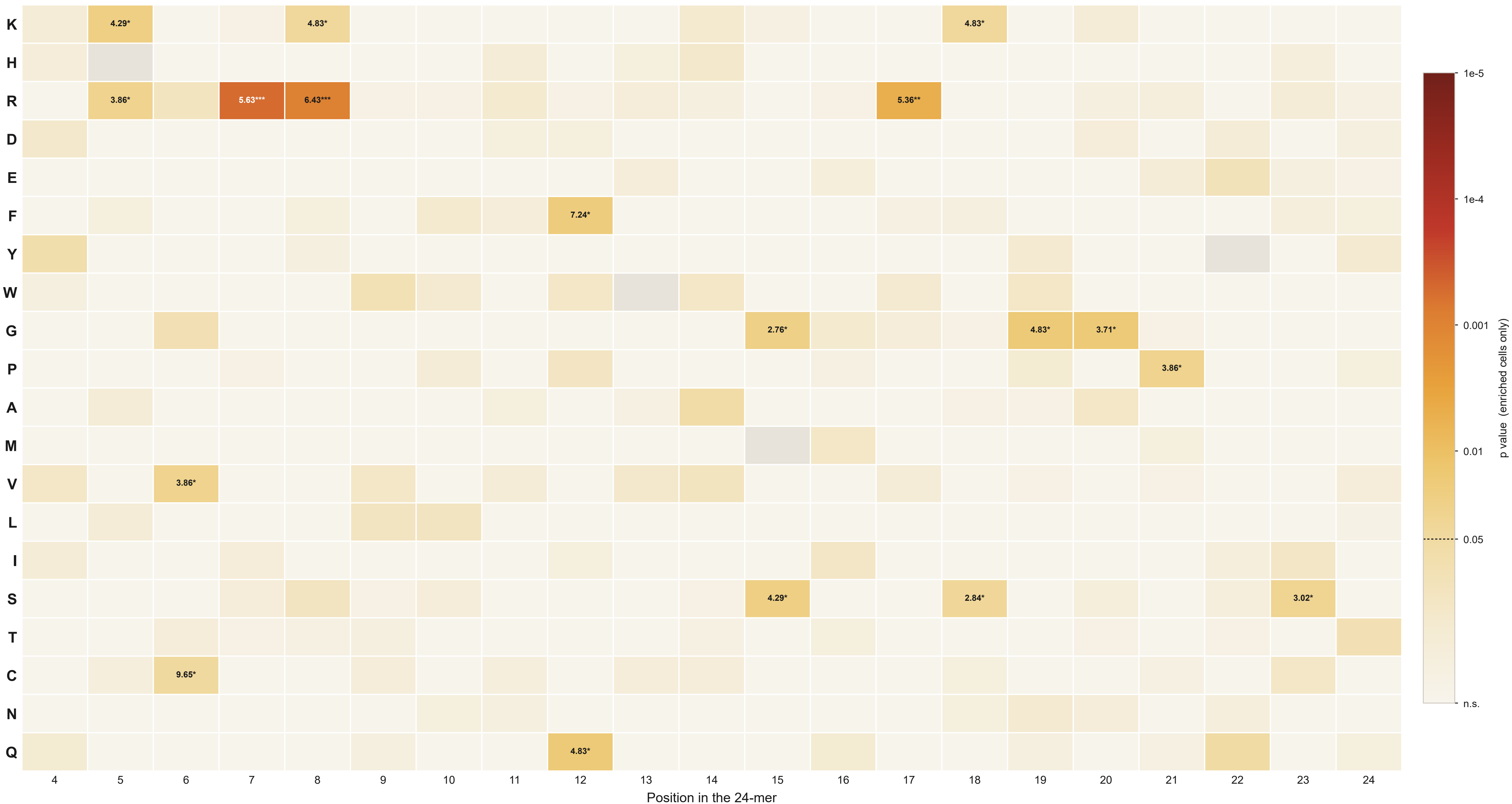
Position × residue enrichment, coloured by significance (enriched cells only)

same layout as the original Colab heatmap, but the colour scale is $-\log_{10}$ p (Fisher exact) and depleted cells are held at the floor



n = 20 P/G-D/E/T substrates vs n = 193 non-substrate controls carrying the same motif. Rows are ordered by chemical class — Basic K H R · Acidic D E · Aromatic F Y W · Aliphatic G P A M · Hydrophobic V L I · Polar S T C N Q.

Cell text is the fold change (substrates / controls) of every cell at $p < 0.05$, followed by the star bin of its uncorrected p: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Grey = residue absent from both groups, no test possible.

No depleted cell reaches $p < 0.05$ anywhere in this dataset, so every coloured cell above is an enrichment. 18 of 416 testable cells reach $p < 0.05$ uncorrected, against 21 expected by chance alone — and none survives FDR correction, so no single cell here is more than a multiple-testing artefact.